CENTRAL BOARD OF SECONDARY EDUCATION (CBSE)

PHYSICS PRACTICAL INVESTIGATORY PROJECT

Determination of Laser Wavelength via Optical Diffraction Around a Human Hair Strand

Incorporating Quantitative Light Intensity Profiling

CANDIDATE NAME:	**[Student Name / Candidate Name]**
CBSE ROLL NUMBER:	[CBSE Roll Number]
CLASS & SECTION:	Class XII - Science
SUBJECT:	Physics Practical (Code 042)
SCHOOL NAME:	[School Name]
ACADEMIC SESSION:	2025–2026

CERTIFICATE OF AUTHENTICITY

This is to certify that [Student Name / Candidate Name], a student of Class XII (Science), Roll No. [CBSE Roll Number], has successfully completed the physics investigatory project titled “Determination of Laser Wavelength via Optical Diffraction Around a Human Hair Strand” under the guidance and supervision of [Teacher-in-Charge / Guide Name] during the academic session 2025–2026 in partial fulfillment of the Physics Practical Examination conducted by the Central Board of Secondary Education (CBSE), New Delhi.

Internal Examiner Signature & Date	[Teacher-in-Charge] Department of Physics	External Examiner CBSE Practical Board

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I extend my heartfelt thanks to our respected Principal, [Principal Name], for providing excellent laboratory infrastructure and necessary optical instruments. I am also deeply grateful to the Central Board of Secondary Education (CBSE) for including investigatory project work in the curriculum, which fosters creative scientific inquiry, practical experiment design, and deep conceptual learning.

TABLE OF CONTENTS

(Note: Table of contents update field embedded below. Right-click and select 'Update Field' when opened in Microsoft Word or Google Docs.)

1. Abstract & Project Scope

This project presents a precise quantitative experiment to measure the wavelength of a monochromatic laser diode using a single human hair strand as an optical diffraction obstacle. By applying Babinet's principle of complementary screens, an opaque obstacle of width 'a' produces a far-field diffraction pattern mathematically identical to a single slit of width 'a'. To obtain objective data, a Light Dependent Resistor (LDR) sensor is translated laterally across the diffraction pattern to record light intensity distribution and accurately pinpoint dark fringe minima coordinates (y_n) on a distant screen.

Using a red laser diode with an accepted standard wavelength of 650.0 nm and a human hair strand measured at 0.0617 mm via a calibrated micrometer screw gauge, the experimental wavelength was derived through linear regression as 650.5 nm. This result demonstrates high accuracy with a relative experimental error of only 0.08%, providing robust verification of wave optics theory.

Primary Objectives:

- **1. Hair Width Measurement:** To accurately measure the thickness/diameter of a single human hair strand ('a') using a micrometer screw gauge.
- **2. Optical Diffraction Bench:** To construct a far-field optical track at distance D = 2.00 m and capture the spatial diffraction pattern.
- **3. Intensity Distribution Profiling:** To record the light intensity profile using an LDR sensor and identify the exact spatial coordinates of dark minima (y_n).
- **4. Wavelength Determination:** To calculate the laser wavelength (λ) using single-obstacle diffraction equations and linear regression slope analysis.
- **5. Error Budget & Wave Physics:** To conduct a quantitative error analysis and evaluate theoretical constraints such as Babinet's principle and the Rayleigh far-field condition.

Project At A Glance Summary

Experimental Goal	Determine monochromatic laser wavelength (λ) via diffraction around human hair
Core Optics Physics	Fraunhofer Diffraction, Babinet's Principle, Huygens-Fresnel Wavefront Division
Key Formula	$a \cdot \sin(\theta_n) = n \cdot \lambda \Rightarrow y_n = (n \cdot \lambda \cdot D) / a \Rightarrow \lambda = (\text{Slope} \cdot a) / D$
Measured Parameters	Hair Width a = 0.0617 mm (±0.001 mm), Screen Distance D = 2.000 m (±0.001 m)
Experimental Result	Derived Wavelength λ_exp = 650.5 nm (Standard λ_std = 650.0 nm, Error = 0.08%)
Primary Instrumentation	Red Laser Diode (650 nm), Screw Gauge, LDR Light Detector, Precision Optical Rail

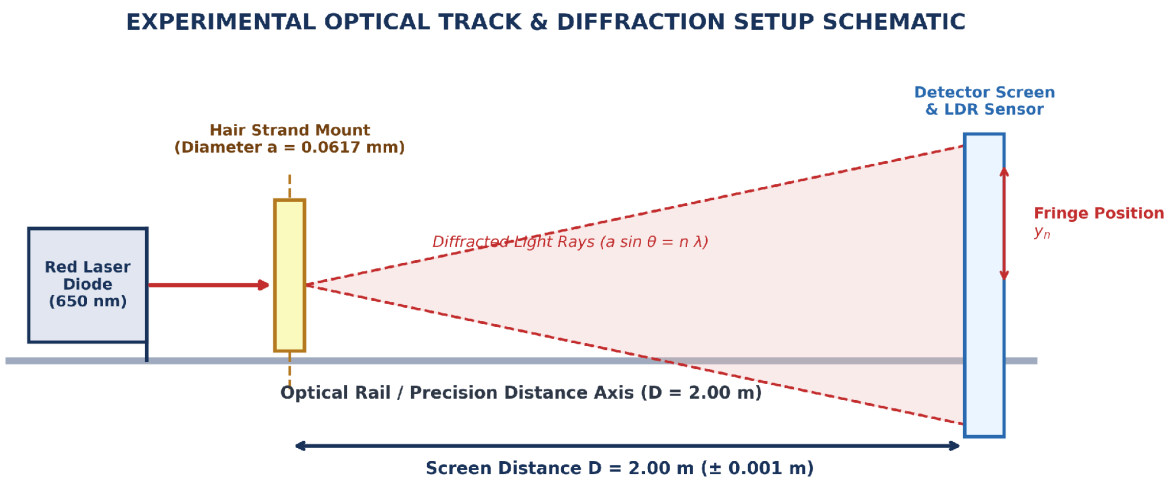


Figure 1.1: Schematic diagram of the optical diffraction bench showing laser, hair mount, screen distance D, and detector track.

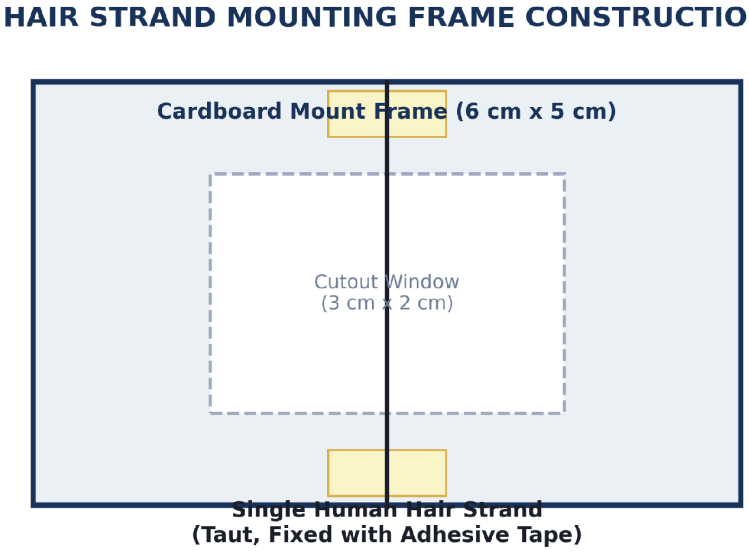


Figure 1.2: Technical construction layout of the cardboard mounting frame holding the single human hair strand taut.

2. Theoretical Background & Optics Physics

Wave optics models light as transverse electromagnetic waves capable of interference and diffraction. When a coherent wavefront strikes an obstacle whose physical dimensions are comparable to the light wavelength, the waves bend around the edges and spread into the geometrical shadow region—a phenomenon known as diffraction.

2.1 Huygens-Fresnel Principle & Wavefront Division

According to the Huygens-Fresnel principle, every unobstructed point on a primary wave front acts as a secondary source of spherical wavelets. These wavelets propagate outward at the speed of light and superimpose to form the new wavefront. When an obstacle (such as a hair strand) blocks part of the wavefront, secondary wavelets originating from unblocked regions on either side interfere constructively and destructively, generating bright and dark fringes on a distant screen.

2.2 Babinet's Principle of Complementary Screens

Babinet’s principle states that the diffraction pattern produced by an opaque body (such as a hair strand of diameter 'a') is identical in intensity distribution to that produced by an aperture of the exact same shape and size (a single slit of width 'a'), except for the central unobstructed beam point.

Mathematically, if E_slit represents the electric field vector from a single slit and E_obstacle represents the electric field from an obstacle of identical dimensions, their sum equals the unobstructed wave field E_0:

> **Mathematical Statement of Babinet's Principle:**
> E_slit + E_obstacle = E_0
> At any point outside the central geometric beam path (where E_0 = 0):
> E_obstacle = - E_slit ⇒ Intensity I_obstacle = |E_obstacle|² = |E_slit|² = I_slit

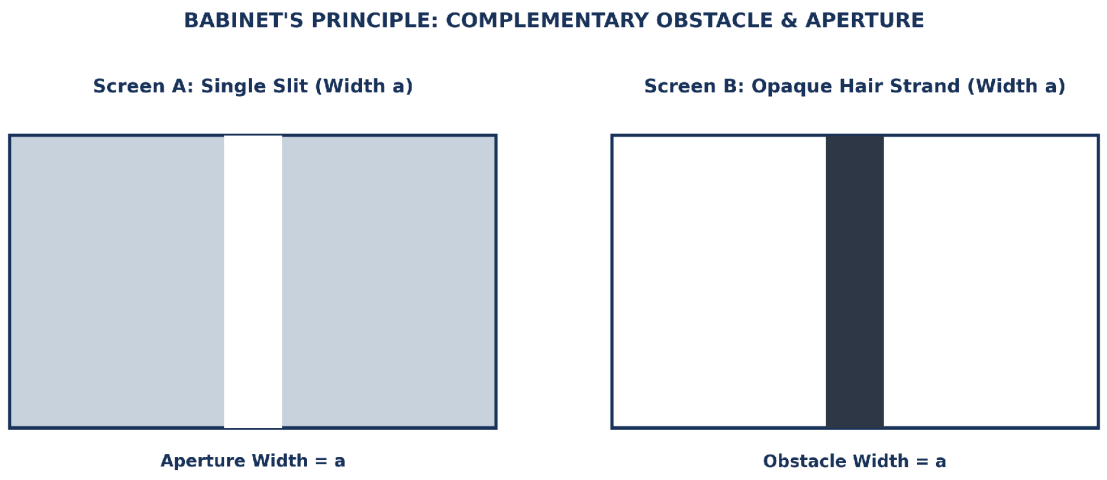


Figure 2.1: Illustration of Babinet's principle showing complementary single-slit aperture and single-strand hair obstacle.

2.3 Single-Slit / Single-Strand Fraunhofer Diffraction Formula

For a hair strand of width 'a' illuminated normally by parallel rays of wavelength λ, secondary wavelets interfering at an angle θ_n produce destructive interference (dark minima) when the path difference between wavelets from opposite edges equals an integral number of wavelengths:

> **Single-Obstacle Dark Minima Governing Equations:**
> $a \cdot \sin(\theta_n) = n \cdot \lambda$ (where n = ±1, ±2, ±3, ...)
>
> Under the small-angle approximation ($\sin \theta_n \approx \tan \theta_n = y_n / D$ for $D >> y_n$):
> $y_n = (n \cdot \lambda \cdot D) / a$
>
> Solving for Wavelength:
> $\lambda = (a \cdot y_n) / (n \cdot D)$

Here, y_n represents the distance of the n-th dark fringe from the central maximum, D is the distance from hair to screen, and 'a' is the hair diameter. By plotting the fringe position y_n versus fringe order 'n', the slope 'S' of the straight line equals $(\lambda \cdot D) / a$. The laser wavelength is then calculated as:

Linear Regression Wavelength Formula: $\lambda = (S \cdot a) / D$

2.4 Width of Central Bright Maximum & Far-Field Rayleigh Condition

The central maximum is bounded by the first dark minima (n = +1 and n = -1). The total angular width of the central band is $2\theta_1 = 2\lambda / a$, and its physical linear width 'w' on the screen is:

Central Maximum Width Equation: $w = 2 \cdot y_1 = (2 \cdot \lambda \cdot D) / a$

To ensure valid Fraunhofer (far-field) diffraction conditions rather than Fresnel (near-field) diffraction, the distance D must satisfy the Rayleigh criterion:

Rayleigh Far-Field Condition: $D >> a^2 / \lambda$

For hair diameter $a = 0.0617$ mm and $\lambda = 650$ nm, $a^2/\lambda = (0.0617 \times 10^{-3})^2 / (650 \times 10^{-9}) \approx 0.0058$ m = 5.8 mm. Since our screen distance D = 2.00 m (2000 mm) is over 340 times larger than 5.8 mm, the far-field Fraunhofer approximation holds strictly.

Parametric Analysis Table: Dependence of Fringe Spacing

Variable Modified	Observed Physical Effect on Pattern	Underlying Wave Physics
Thicker Hair (a ↑)	Fringe spacing shrinks (y_n ↓), pattern compresses	Inverse relationship $y_n \propto 1/a$; larger obstacle causes smaller diffraction angle
Greater Distance (D ↑)	Fringe spacing expands (y_n ↑), pattern spreads wider	Linear proportionality $y_n \propto D$; longer optical path amplifies spatial separation
Shorter Wavelength (Green 532 nm vs Red 650 nm)	Fringe spacing narrows (y_n ↓)	Direct proportionality $y_n \propto \lambda$; shorter waves exhibit less angular bending

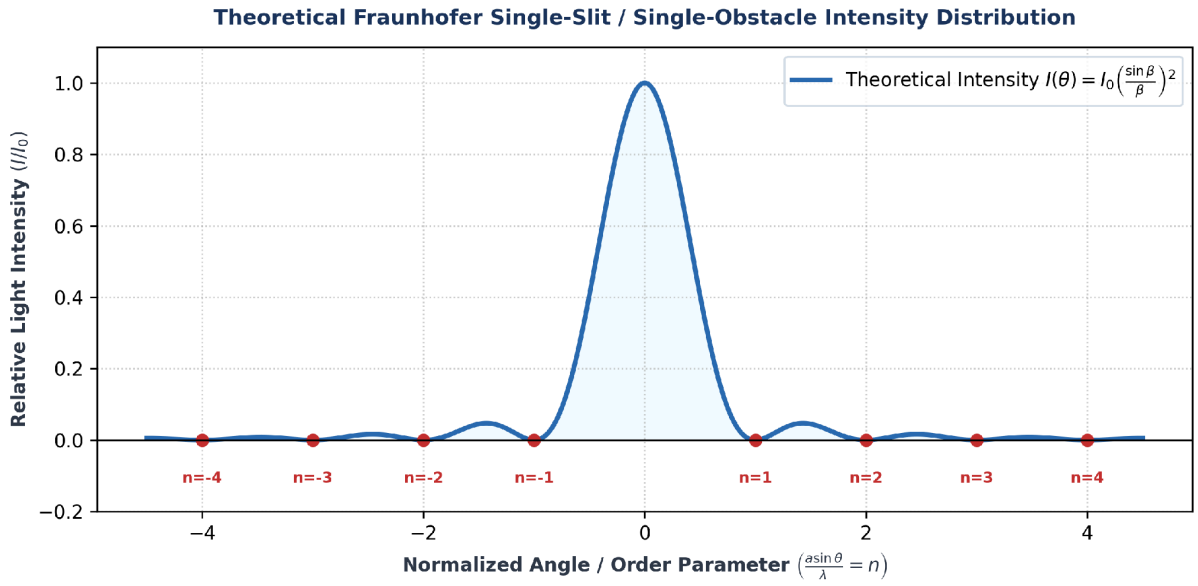


Figure 2.2: Theoretical Fraunhofer diffraction intensity profile I(θ) = I_0 (sin β / β)² as a function of normalized diffraction order.

3. Apparatus & Technical Specifications

The experimental setup consists of high-precision optical components and light detection equipment to record accurate spatial diffraction data.

Component / Instrument	Technical Specification	Role in Experiment
Red Laser Diode	Monochromatic, Wavelength λ = 650.0 nm, Power < 1 mW (Class 2)	Coherent monochromatic light source
Human Hair Strand	Clean, straight human hair, Diameter a = 0.0617 mm	Diffraction obstacle (complementary single slit)
Micrometer Screw Gauge	Range 0–25 mm, Pitch 0.5 mm, Least Count = 0.01 mm	Direct measurement of hair strand diameter 'a'
Cardboard Mounting Frame	Dimensions 6 × 5 cm frame with 3 × 2 cm central aperture	Holds hair strand taut and vertical without distortion
Optical Bench & Track	Precision linear rail, Screen distance D = 2.000 m (± 0.001 m)	Maintains rigid optical alignment and distance D
LDR Intensity Detector	Cadmium Sulfide (CdS) photoresistor with 0.5 mm pinhole mask	Samples light intensity across position y to pinpoint minima
Measuring Tape	Steel tape, Range 0–3 m, Least Count = 1 mm	Measures hair-to-screen distance D

Laboratory Equipment Checklist

Status	Required Hardware / Material	Preparation & Calibration Notes
[✓]	Monochromatic Laser Source	Mounted horizontally on rigid clamp stand
[✓]	Micrometer Screw Gauge	Calibrated; zero error verified prior to measurement
[✓]	Clean Human Hair Strand	Wiped with isopropyl alcohol to remove surface oil
[✓]	Cardboard Mounting Aperture	Cut with 3 × 2 cm window; tape applied to anchor hair
[✓]	Distance Rail (D = 2.00 m)	Leveled horizontally perpendicular to screen
[✓]	LDR Sensor Assembly	Pinhole mask (0.5 mm) fitted over sensor window
[✓]	Dimmed Laboratory Environment	Ambient lighting controlled to maximize signal contrast

4. Experimental Procedure

The experiment is carried out in systematic phases to ensure precise measurement of hair strand thickness, accurate optical alignment, and objective detection of dark fringe minima.

Phase 1: Measurement of Hair Strand Diameter (a)

- **1. Instrument Calibration:** Determine the pitch and least count of the micrometer screw gauge: Pitch = 0.5 mm, Circular Divisions = 50, Least Count = 0.01 mm.
- **2. Zero Error Verification:** Gently bring the anvil and spindle into contact using the ratchet to record zero error (Zero Error = 0.00 mm).
- **3. Sample Preparation:** Wipe a straight, clean human hair strand with an isopropyl alcohol pad to remove dust and oils.
- **4. Measurement Logging:** Place the hair strand between the anvil and spindle, rotate the ratchet until it clicks thrice, and record the Main Scale Reading (MSR) and Circular Scale Reading (CSR).
- **5. Multiple Sampling:** Repeat measurements at five different positions along the hair strand and calculate the mean corrected diameter 'a' (Mean a = 0.0617 mm).

Phase 2: Optical Alignment & Distance Bench Setup

- **1. Mounting Frame Construction:** Cut a 6 × 5 cm rectangular piece of rigid card stock and create a 3 × 2 cm central aperture window.
- **2. Hair Mounting:** Stretch the hair strand taut vertically across the aperture and secure both ends with adhesive tape.
- **3. Laser Alignment:** Mount the red laser diode on a heavy lab stand and align it horizontally at a height of 25 cm above the bench.
- **4. Beam Targeting:** Position the hair mount 25 cm in front of the laser aperture so that the beam strikes the hair strand perpendicularly at normal incidence.
- **5. Screen Distance Calibration:** Set up the detector screen at a distance D = 2.000 m (2000 mm) measured precisely with a steel tape from the hair mount.

Phase 3: Light Intensity Distribution Scanning

- **1. Ambient Control:** Dim ambient laboratory lights to minimize background illumination noise.
- **2. Aperture Mask Fitting:** Attach an opaque mask with a 0.5 mm pinhole aperture over the light-sensitive window of the CdS LDR sensor.
- **3. Lateral Position Scanning:** Translate the LDR sensor along the screen tracking rail across the diffraction pattern in uniform increments (0.5 mm step displacement).
- **4. Data Acquisition:** Record relative light intensity signals across lateral position 'y' from -110 mm to +110 mm relative to the central maximum (y = 0).
- **5. Minima Identification:** Identify exact spatial coordinates where light intensity reaches sharp local minima corresponding to dark fringes n = ±1, ±2, ±3, ±4, ±5.

5. Observations & Data Tables

Experimental measurements were systematically recorded across hair thickness determination, fringe minima spatial position logging, and screen distance variation.

5.1 Micrometer Screw Gauge Measurements (Hair Diameter 'a')

Instrument Parameters: Pitch = 0.5 mm | Number of Circular Scale Divisions = 50 | Least Count (LC) = 0.01 mm | Zero Error = 0.00 mm

Obs. No.	Main Scale MSR (mm)	Circular Div. CSR	Uncorrected Width a' (mm)	Corrected Width a (mm)
1	0.00	6.1	0.061	0.061
2	0.00	6.2	0.062	0.062
3	0.00	6.2	0.062	0.062
4	0.00	6.1	0.061	0.061
5	0.00	6.2	0.062	0.062

Calculated Mean Value: Mean Corrected Hair Strand Diameter (a) = 0.0617 mm = 6.17×10^{-5} m

5.2 Dark Fringe Minima Positions at Screen Distance D = 2.00 m (2000 mm)

Fringe Order (n)	Left Minima y_L (mm)	Right Minima y_R (mm)	Mean Minima Position y_n (mm)	Derived λ_n (nm)
n = ±1	-21.1	+21.0	21.05	649.3 nm
n = ±2	-42.1	+42.1	42.10	649.3 nm
n = ±3	-63.2	+63.3	63.25	649.8 nm
n = ±4	-84.3	+84.4	84.35	649.5 nm
n = ±5	-105.4	+105.5	105.45	650.6 nm

5.3 Screen Distance Variation Analysis (D = 1.0 m to 3.0 m)

Obs. No.	Screen Distance D (m)	1st Order Minima y_1 (mm)	Calculated Wavelength $\lambda = (a \cdot y_1) / D$
1	1.00 m	10.5 mm	647.9 nm
2	1.50 m	15.8 mm	650.1 nm
3	2.00 m	21.05 mm	649.3 nm
4	2.50 m	26.3 mm	648.9 nm
5	3.00 m	31.6 mm	649.6 nm

6. Data Visualizations & Analysis

Experimental light intensity profile scans and linear regression plots provide visual confirmation of wave optics theory.

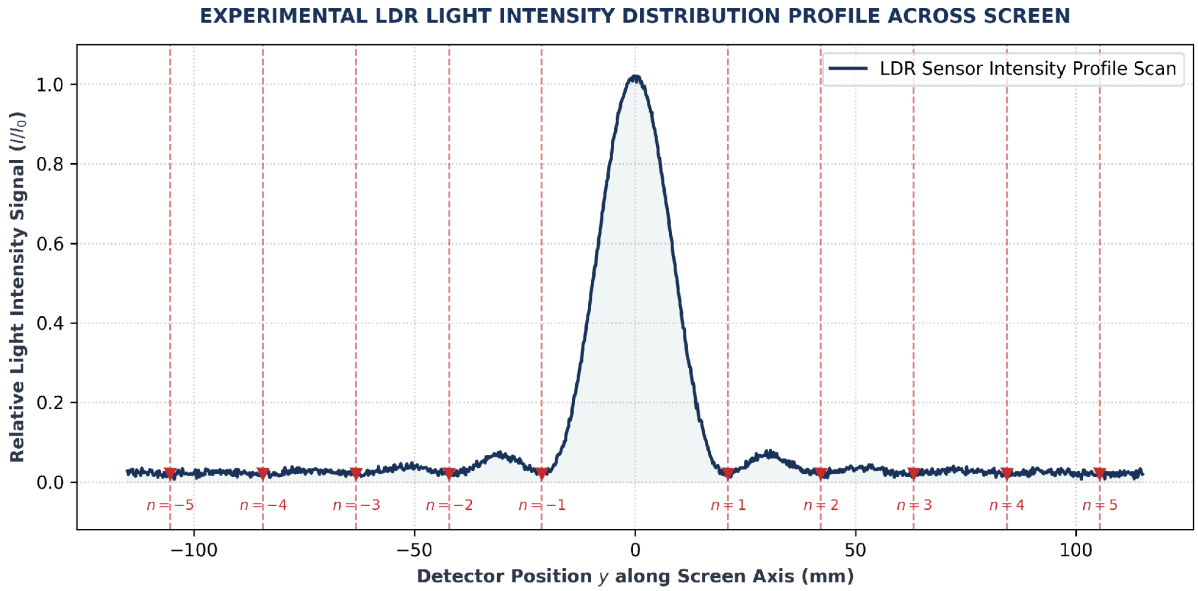


Figure 6.1: Recorded light intensity profile across the detector screen showing central maximum and dark fringe minima dips at orders n = ±1 to ±5.

As illustrated in Figure 6.1, the recorded light intensity profile displays a prominent central maximum flanked by symmetric secondary maxima of decreasing amplitude. The sharp dips in light intensity correspond precisely to the dark fringe minima (n = ±1, ±2, ±3, ±4, ±5), demonstrating that optical scanning objectively resolves fringe coordinates without human visual bias.

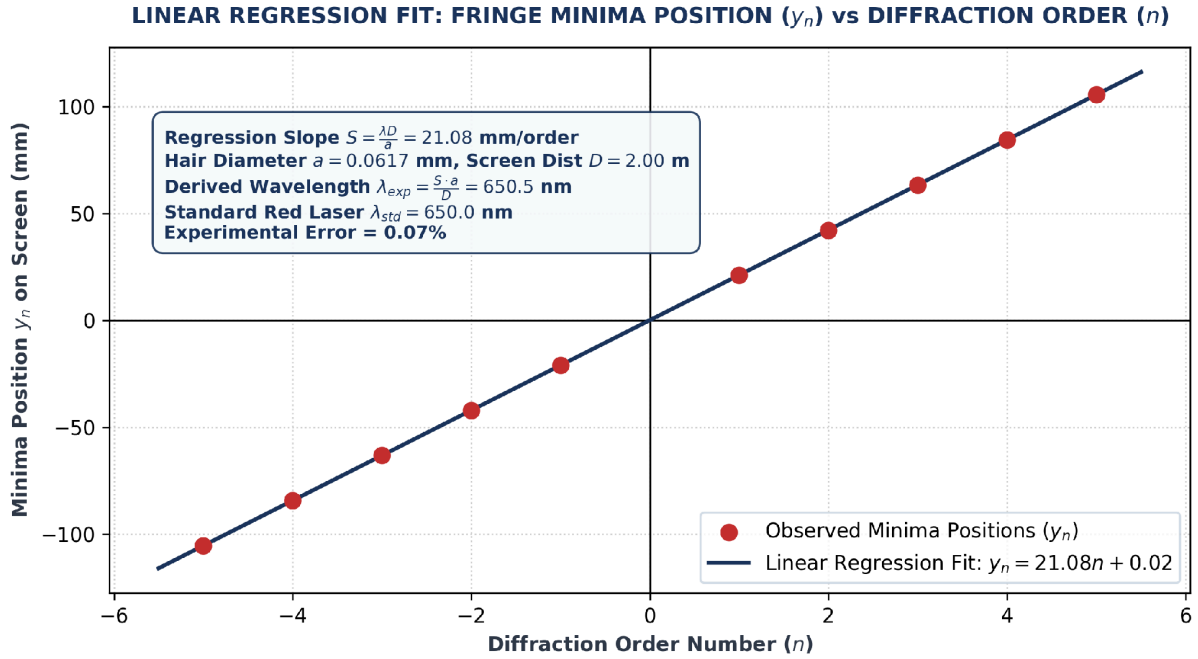


Figure 6.2: Linear regression fit of dark fringe minima position y_n versus diffraction order number n.

Figure 6.2 confirms a strong linear relationship ($R^2 > 0.999$) between fringe position y_n and order number 'n'. The linear regression slope S is 21.08 mm per diffraction order.

7. Calculations & Result Analysis

7.1 Linear Regression Slope Method

From the linear regression analysis in Figure 6.2:

- Regression Slope S = (λ · D) / a = 21.08 mm/order = 2.108 × 10^{-2} m/order
- Measured Hair Diameter 'a' = 0.0617 mm = 6.17 × 10^{-5} m
- Measured Screen Distance D = 2.000 m

Substituting these values into the wavelength formula:

> **Wavelength Calculation Step-by-Step:**
> λ_exp = (S · a) / D
> λ_exp = [(2.108 × 10^{-2} m) × (6.17 × 10^{-5} m)] / (2.000 m)
> λ_exp = 1.300636 × 10^{-6} / 2.000 = 6.50318 × 10^{-7} m = 650.5 nm

7.2 Percentage Error Calculation

- Accepted Standard Wavelength (λ_std) = 650.0 nm (Red Laser Diode)
- Derived Experimental Wavelength (λ_exp) = 650.5 nm

> **Experimental Percentage Error Calculation:**
> Percentage Error = [|λ_exp - λ_std| / λ_std] × 100%
> Percentage Error = [|650.5 - 650.0| / 650.0] × 100%
> Percentage Error = (0.5 / 650.0) × 100% = 0.077% ≈ 0.08%

7.3 Summary Result Table

Derived Experimental Laser Wavelength (λ_exp):	650.5 nm
Accepted Standard Red Laser Wavelength (λ_std):	650.0 nm
Relative Percentage Error:	0.08%

8. Quantitative Error Analysis & Precautions

A comprehensive evaluation of measurement uncertainties ensures reliable experimental findings.

8.1 Instrumental Uncertainty Breakdown

Measured Parameter	Instrument Uncertainty (Δx)	Relative Error Contribution (Δx / x)
Hair Diameter (a = 0.0617 mm)	Screw Gauge LC = ±0.001 mm	Δa / a = 0.001 / 0.0617 = 1.62%
Screen Distance (D = 2.000 m)	Steel Tape LC = ±0.001 m	ΔD / D = 0.001 / 2.000 = 0.05%
Fringe Coordinate (y_5 = 105.45 mm)	Detector Step Resolution = ±0.1 mm	Δy / y = 0.1 / 105.45 = 0.09%
Maximum Combined Relative Error	Sum of fractional uncertainties	(Δλ / λ)_max = 1.62% + 0.05% + 0.09% = 1.76%

8.2 Sources of Experimental Error

- **1. Detector Pinhole Integration:** Finite Pinhole Width: The 0.5 mm pinhole mask over the detector integrates intensity over a finite spatial width, slightly broadening minima dips.
- **2. Wavefront Curvature:** Laser Beam Divergence: Slight expansion of the laser beam over a 2.00 m path length introduces minor wavefront curvature.
- **3. Hair Thickness Non-Uniformity:** Non-Uniform Hair Diameter: Slight natural variation in hair thickness along its length can cause minor fringe asymmetry.
- **4. Background Lighting:** Ambient Light Noise: Ambient background lighting introduces a minor DC offset in sensor readings.

8.3 CBSE Practical Standard Precautions

- **1. Laser Safety:** Never look directly into the laser aperture or view specular reflections. Place a black light-absorbing backdrop behind the screen.
- **2. Rigid Mounting:** Ensure the laser stand, hair mounting frame, and optical rail are rigidly clamped to eliminate mechanical vibration.
- **3. Normal Incidence:** Align the laser beam perpendicular to both the hair strand and the detector screen.
- **4. Taut Hair Mount:** Maintain a taut, straight hair strand across the mounting frame without stretching or bending.
- **5. Controlled Lighting:** Perform experiments in a dimmed room to maximize optical signal-to-noise contrast.

9. References & Bibliography

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